Spencer Lee

Curriculum Vitae

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Experience

2021–Present **PhD Researcher**, *Michigan State University*, East Lansing, MI.

- Created and maintain QuantumGateDesign.jl, an open-source Julia package (CI tests, hosted docs) for fast and accurate pulse-level optimal control of superconducting transmon qubits.
- Demonstrated 3–13× faster CNOT gate design than Störmer-Verlet at 10^{-3}–10^{-5} infidelity targets, using the high-order Hermite methods and discrete adjoints of QuantumGateDesign.jl (*Journal of Computational Physics, 2026*).
- Derived Filon methods that simulate highly oscillatory quantum systems 6× faster than same-order Hermite methods, with no rotating-wave approximation (arXiv:2607.13580; submitted to *SIAM Journal on Scientific Computing*).
- Built a transfer-learning pipeline, from dataset generation through model training, that retunes control pulses to preserve gate fidelity on qubits with shifted transition frequencies.
- Currently designing neural-network surrogate models for controlled qubit dynamics using learned Magnus expansions, enabling state-vector simulation with large time steps at low computational cost.

2023, 2025 **Long Workshop Participant**, *Institute for Pure and Applied Mathematics*, Los Angeles, CA.

- Collaborated with a diverse group of mathematicians, physicists, and computer scientists from academia and industry on problems in quantum computing (2023) and non-commutative optimal transport (2025).
- Organized a working group exploring optimal-transport-inspired objective functions for quantum control.

2024 **Research Project Manager**, *Tohoku University (G-RIPS Program)*, Sendai, Japan.

- Implemented GPU-accelerated algorithms for classical parameter-setting of the Quantum Approximate Optimization Algorithm (QAOA), in collaboration with Mitsubishi Electric.
- Led a five-member U.S.–Japan team that replaced a state-of-the-art parameter-setting method with a low-cost analytical proxy of comparable accuracy.

2023 **Visiting Scholar**, *Los Alamos National Laboratory*, Los Alamos, NM.

- Wrote CUDA and Thrust code to accelerate simulation of runaway electrons in tokamak fusion reactors.
- Visualized electron dynamics using Poincaré plots to identify computationally demanding reactor regions.

2022 **Computing Student Intern**, *Lawrence Livermore National Laboratory*, Livermore, CA.

- Calibrated gate pulses and implemented zero-noise extrapolation on real superconducting qubit hardware.
- Measured hardware coherence and accuracy via Ramsey echo and randomized benchmarking experiments.
- Developed high-order compositional integrators for quantum state-vector simulation, which reduced the computational cost of high-accuracy 2-qubit simulations by 10× (*Lawrence Livermore National Laboratory Technical Report, 2022*).

2020–2021 **Post-Baccalaureate Research Associate**, *Michigan State University*, East Lansing, MI.

- Member of the FEX-Hub project for development of hardware to manage the flow of trigger data from the ATLAS detector of the Large Hadron Collider.
- Wrote object-oriented Python scripts to conduct Integrated Bit Error Ratio Tests (IBERT) of multi-gigabit transceivers on FPGA and generate eye diagrams to analyze accuracy when transmitting at high bandwidths.
- Added new features to FPGA firmware using VHDL in Vivado Design Suite.

2019 **Summer Research Associate**, *CERN*, Geneva, Switzerland.

- Analyzed C firmware of a microcontroller for ATCA compliance as part of upgrades to the Level 1 Calorimeter Trigger of the Large Hadron Collider.

2016–2020 **Undergraduate Research Associate**, *Michigan State University*, East Lansing, MI.

- Created an interactive Python GUI to run and display perfSONAR network measurement tests, manage Ethernet switch stress tests, and generate automated reports.
- Organized and configured Linux test nodes for use in network stress tests.

Education

2021–Present **Ph.D. in Computational Mathematics, Science, & Engineering**, *Michigan State University*, East Lansing, MI, *GPA – 3.94/4.0*.
Expected graduation October 2026.

2016–2020 **Dual Bachelor of Science in Advanced Mathematics and Physics**, *Michigan State University*, East Lansing, MI, *GPA – 3.97/4.0*.
Graduated with High Honors, Member of Honors College.

Awards

2022–2027 **National Science Foundation Graduate Research Fellowship** ($159,000)

2021 Michigan State University Rasmussen Fellowship Award ($5,000)

2019 McCartney Endowed Educational Enrichment Fund ($500)

2018 Lawrence W.Hantel Endowed Fellowship Fund ($2,300)

2018 Paul and Wilma Dressel Endowed Scholarship for Mathematics ($1,250)

Publications

Published Articles

2026 Spencer Lee and Daniel Appelö. *High-order Hermite optimization: Fast and exact gradient computation in open-loop quantum optimal control using a discrete adjoint approach*. Journal of Computational Physics*, 552, p. 114697.* [Link]

Submitted Articles

2026 Spencer Lee and Daniel Appelö. *Filon Methods for Highly Oscillatory Controlled Quantum Systems*. arXiv:2607.13580, submitted to the SIAM Journal on Scientific Computing. [Link]

Technical Reports & White Papers

2025 White paper: *Non-commutative Optimal Transport*. *Institute for Pure and Applied Mathematics (IPAM), UCLA*. [Link]

2024 Rie Fujii et al. Technical Report: *G-RIPS Mitsubishi-B Project: Final Report*. *Tohoku University Advanced Institute for Materials Research (AIMR)*. [Link]

2024 White paper: *Mathematical and Computational Challenges in Quantum Computing*. *Institute for Pure and Applied Mathematics (IPAM), UCLA*. [Link]

2022 Spencer Lee, Stefanie Guenther, and N. Anders Petersson. Technical Report: *Compositional Methods for Schrödinger's Equation with Application to Optimal Control*. *Lawrence Livermore National Laboratory (LLNL)*. [Link]

2018 Dan Edmunds et al. Technical Specification: *ATLAS Level-1 Calorimeter Trigger Update, HUB Firmware Specification*. *Michigan State University*.

2018 Dan Edmunds et al. Technical Specification: *ATLAS Level-1 Calorimeter Trigger Upgrade, FEX System Switch Module (FEX Hub) Prototype*. *Michigan State University*.

Talks

2025 *Building Blocks of Quantum Computing: QuantumGateDesign.jl - Quantum Optimal Control Tutorial*. Talk at Society for Industrial and Applied Mathematics (SIAM) Annual Meeting 2025, Montreal, Quebec.

2025 *High-Order Hermite Optimization for Quantum Gate Design*. Talk at Society for Industrial and Applied Mathematics (SIAM) Annual Meeting 2025, Montreal, Quebec.

2025 *HOHO - High-Order Hermite Optimization for Quantum Optimal Control*. Talk at CMSE 10th Anniversary Workshop & 5th Data Science Student Conference, Michigan State University, East Lansing, Michigan.

2025 *Quantum Optimal Control, Barren Plateaus, and Wasserstein Distances*. Talk at Institute for Pure and Applied Mathematics (IPAM) NOT2025 Culminating Retreat, Lake Arrowhead, California.

2025 *Quantum Optimal Control: How It's Done and Why You Should Use My Software*. Talk at Institute for Pure and Applied Mathematics (IPAM) CQC2023 Reunion, Lake Arrowhead, California.

2024 *Classical Parameter-Setting Strategies for the Quantum Approximate Optimization Algorithm (QAOA).* Talk at Graduate-level Research in Industrial Projects for Students (GRIPS) Final Presentation, Tohoku University Advanced Institute for Materials Research (AIMR), Sendai, Japan.

2024 *Designing Quantum Gates Using High-Order Hermite Methods.* Talk at Society for Industrial and Applied Mathematics (SIAM) Annual Meeting 2024, Online.

2024 *Fast, High-Order Solvers for Quantum Optimal Control.* Talk at Quantum Information and Computation (QuIC) Seminar, Michigan State University, East Lansing, Michigan.

2023 *Classically-Simulated Quantum Optimal Control Methods for Designing Gates using Continuous Control Functions.* Talk at Institute for Pure and Applied Mathematics (IPAM) CQC2023 Student Seminar Series, IPAM, Los Angeles, California.

2023 *Implicit Filon Methods for Highly Oscillating Problems, Gate Design for Controlled Qubits.* Talk at American Mathematical Society (AMS) 2023 Spring Sectional, University of Cincinnati, Cincinnati, Ohio.

2022 *Implicit Filon Methods for Highly Oscillating Problems, Gate Design for Controlled Qubits.* Talk at Waves 2022, ENSTA-Paris, Palaiseau, France.

Posters

2025 *QuantumGateDesign.jl: High-Order Methods, Fast Gate Design.* Poster at International conference on Quantum Simulation and Quantum Walks (QSQW) 2025, Naples, Italy.

2024 *QuantumGateDesign.jl: Higher-Order Methods for Faster Gate Design.* Poster at Kernel Methods in Uncertainty Quantification and Experimental Design workshop, Institute for Mathematical and Statistical Innovation, Chicago, Illinois.

2023 *Filon Time Integration for Gate Design.* Poster at Accelerated Research in Quantum Computing (ARQC) All-hands Meeting, Berkeley National Lab, Berkeley, California.

2022 *Numerical Methods for Highly Oscillatory Problems in Quantum Computing: Quantum State Evolution Simulated using Implicit Filon Quadrature.* Poster at Midwest Numerical Analysis Day 2022, University of Michigan, Ann Arbor, Michigan.

Skills

Programming: Julia, Python, C, C++, VHDL, Matlab, Mathematica

HPC & Sci. Computing: CUDA, MPI, OpenMP, SLURM, Adjoint Methods, High-Order Methods, Profiling

Quantum Computing: QuTiP, Qiskit, CUDA-Q, Pulse-Level Control, QAOA, Error Mitigation

Machine Learning: TensorFlow, Scikit-Learn, Lux.jl, Transfer Learning, Scientific ML

Tools: Git, Linux, Tests & CI, Claude Code, LaTeX

Interests	Quantum Optimal Control, Simulating Physics, Scientific Software, Visualizing Information, Applied Research, Pedagogy, Computer Graphics
Soft Skills	Leadership, Independent Learning, Communication, Presentation, Teamwork